

Chirag Joshi

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Summary

Performance-focused Backend Engineer who reduced MySQL CPU utilization from **97% to 20%** through query optimization and multi-tier caching at **~900 req/s**, and cut CI/CD pipeline duration from **40 minutes to under 5**. GATE CS 2025 qualified (top 5%). Building production systems with Django/FastAPI, Go, Redis, MySQL, and Kubernetes.

Professional Experience

Software Developer

Backend Engineering (Django/DRF, FastAPI, Redis)

InstaAstro

March 2025 – Present

- Reduced MySQL CPU utilization from **97% to 20%** at **~900 req/s** peak traffic by eliminating N+1 queries, introducing multi-tier Redis caching, and restructuring query patterns on multi-million row tables.
- Cut CI/CD pipeline duration from **40 minutes to under 5** by introducing Docker layer caching and migrating from dind to Kaniko.
- Designed and deployed backend for a new domain-specific service as sole contributor, covering API design, authentication logic, and internal admin tooling.
- Optimized multiple APIs (**avg. 35% faster**) by adding targeted database indexes, reducing redundant I/O, and restructuring execution paths to minimize CPU context switching.
- Engineered a production-safe APM middleware stack capturing per-request latency, CPU/memory usage, and per-query SQL durations — used to identify and eliminate slow endpoints across the codebase.

Technical Skills

Languages: C++, Python, Go, JavaScript

Backend: Django, Django REST Framework, FastAPI, PostgreSQL, MySQL, Redis, Celery, Caching

DevOps/Tools: Linux, Docker, Kubernetes, Git, GitHub, GitLab CI/CD

Projects

Real-Time Streaming | Go, Redis, SSE

GitHub

- Built a zero-dependency streaming platform in Go — runs fully in-process with no external services required; plugs into Redis pub/sub for horizontal fan-out when needed.
- Designed a tiered pub/sub abstraction (in-memory → Redis) behind a single interface, enabling single-instance and multi-instance deployments from the same binary with no code changes.
- Delivered real-time chat and viewer-count events over SSE using goroutines and channels, with stream ownership enforced via a secret stream key.

Django TraceKit | Python, Django

GitHub

- Developed a lightweight Django APM library with middleware for per-request response times, CPU/memory profiling, and per-query SQL capture — a minimal, production-safe alternative to django-silk.
- All instrumentation is runtime-configurable via Django admin (URL filters, user IDs, sampling rate, time windows) with a 3-layer cache (request → Redis → DB) — no redeployment needed.
- Implemented a **TimeLogger** helper for per-step wall-clock, CPU, and RSS checkpoints inside views, with automatic caller tracking via **inspect** and batch DB writes.

Education

Maharaja Agrasen Institute of Technology

B.Tech. Computer Science and Engineering (CGPA: **8.32/10**)

New Delhi, India

August 2020 – June 2024

Achievements

- GATE CS 2025:** Qualified, top 5% among 100,000+ candidates.
- LeetCode rating: 1825** (Top 6% globally) | **750+** problems solved across LeetCode/CodeChef/GeeksforGeeks.